

Table 1: Secondary engineering-proxy error differences: capacity-matched DeepSets minus sum GNN; positive values favor the GNN. All differences and interval endpoints are in 10^{-3} response units. End-time error uses the 60-min response; J is the trapezoidal time-average of the 61-point trajectory. Equal-case errors are averaged within each seed and then across five seeds. Intervals use the existing 1,000 crossed geometry-seed resamples and are unadjusted secondary intervals. N counts geometries, not repeated predictions; populations overlap across protocols. These are archival-response proxies, not physical capacity or fire-resistance validation.

Protocol	N	ΔMAE_{60} [95% CI]	ΔMAE_J [95% CI]
IID	160	-0.834 [-1.645, -0.040]	-0.169 [-0.752, 0.414]
Count OOD	264	0.749 [-0.891, 2.230]	0.846 [0.116, 1.607]
ISO 834	158	-0.551 [-1.605, 0.170]	0.242 [-0.226, 0.695]
ASTM E119	95	0.639 [0.030, 1.312]	0.372 [-0.264, 0.977]
Perturbed ISO	111	0.209 [-0.246, 0.700]	0.243 [-0.283, 0.822]
Plateau	103	1.116 [-0.785, 3.864]	0.461 [-0.551, 1.384]
Decay	113	-0.892 [-10.447, 7.663]	-0.415 [-1.969, 1.179]
Log variant	69	2.056 [-0.145, 4.404]	0.357 [-0.961, 1.418]
External fire	49	0.717 [-1.073, 2.783]	-0.161 [-1.348, 1.059]
Bilinear	77	8.162 [-10.552, 34.487]	-0.186 [-7.133, 8.086]
Linear	125	6.728 [-8.265, 23.525]	0.452 [-10.206, 12.063]
Smoldering	97	-1.528 [-24.120, 21.365]	0.518 [-2.641, 3.649]

Table 2: Discrete archival-response screening at $q = 0.8$ and 0.6 . E/C gives reference events/right-censored cases by 60 min, each geometry counted once. Events use the first stored value at or below q on the 60-s grid. Each model reports the separate maximum FN and FP counts over five seeds and its minimum reference-prediction common-event coverage n_{com}/E . These extrema may come from different seeds and are not one worst checkpoint. Common events are model-specific, not a shared two-model cohort. Coverage is NA when $E = 0$; undefined rates and conditional times remain NA in the all-model CSVs. Zero observed errors do not establish zero future risk.

Protocol	q	E/C	Matched set		Sum GNN	
			Max FN/FP	Min n_{com}/E	Max FN/FP	Min n_{com}/E
IID	0.8	144/16	0/0	144/144	0/0	144/144
	0.6	141/19	0/0	141/141	0/0	141/141
Count OOD	0.8	226/38	0/0	226/226	0/0	226/226
	0.6	219/45	1/0	218/219	1/0	218/219
ISO 834	0.8	158/0	0/0	158/158	0/0	158/158
	0.6	158/0	0/0	158/158	0/0	158/158
ASTM E119	0.8	95/0	0/0	95/95	0/0	95/95
	0.6	95/0	0/0	95/95	0/0	95/95
Perturbed ISO	0.8	111/0	0/0	111/111	0/0	111/111
	0.6	111/0	0/0	111/111	0/0	111/111
Plateau	0.8	103/0	0/0	103/103	0/0	103/103
	0.6	103/0	0/0	103/103	0/0	103/103
Decay	0.8	113/0	0/0	113/113	0/0	113/113
	0.6	113/0	0/0	113/113	0/0	113/113
Log variant	0.8	69/0	0/0	69/69	0/0	69/69
	0.6	69/0	0/0	69/69	0/0	69/69
External fire	0.8	49/0	0/0	49/49	0/0	49/49
	0.6	49/0	0/0	49/49	0/0	49/49
Bilinear	0.8	77/0	0/0	77/77	0/0	77/77
	0.6	73/4	0/4	73/73	0/4	73/73
Linear	0.8	122/3	12/0	110/122	12/0	110/122
	0.6	92/33	3/0	89/92	1/2	91/92
Smoldering	0.8	1/96	0/25	1/1	0/25	1/1
	0.6	0/97	0/0	NA	0/0	NA